

Spatial patterns of warming in MPI-ESM1-2-LR (SSP2-4.5)

Comparison of two ensemble members (r1, r25) against recent and preindustrial baselines.

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The four anomaly maps (2091-2100 minus 2016-2025 and minus the full piControl mean, for each ensemble member) show the canonical CMIP6 warming pattern under SSP2-4.5. Warming is everywhere positive and dominated by two well-known features: strong polar amplification over the Arctic (local maxima exceeding 12 K against the preindustrial baseline) and a pronounced land-ocean contrast, with mid- and high-latitude continents warming substantially faster than the surrounding oceans. The Southern Ocean and the North Atlantic 'warming hole' south of Greenland show the slowest warming, consistent with deep ocean heat uptake and a weakening AMOC.

The two experiments (r1, r25) are two realisations of the same forced scenario, so their global-mean response is essentially identical (A: +2.47 K vs B: +2.50 K relative to piControl). They differ, however, in the regional pattern: peak Arctic anomalies in r1 reach 8.6 K against the recent baseline, while r25 peaks at only 6.7 K. This spread is internal-variability noise (decadal sea-ice and Arctic-Oscillation fluctuations), not a forced difference.

The choice of baseline strongly affects the apparent magnitude of the anomaly. The preindustrial baseline yields ~2.5 K of global warming by 2091-2100, whereas the 2016-2025 baseline only shows ~1.3 K, since most of that ~1.1 K of historical and early-21st-century warming has already been absorbed into the recent reference. The recent baseline is therefore the appropriate frame for communicating committed future warming, while the preindustrial baseline is the relevant frame for Paris-Agreement-style targets.

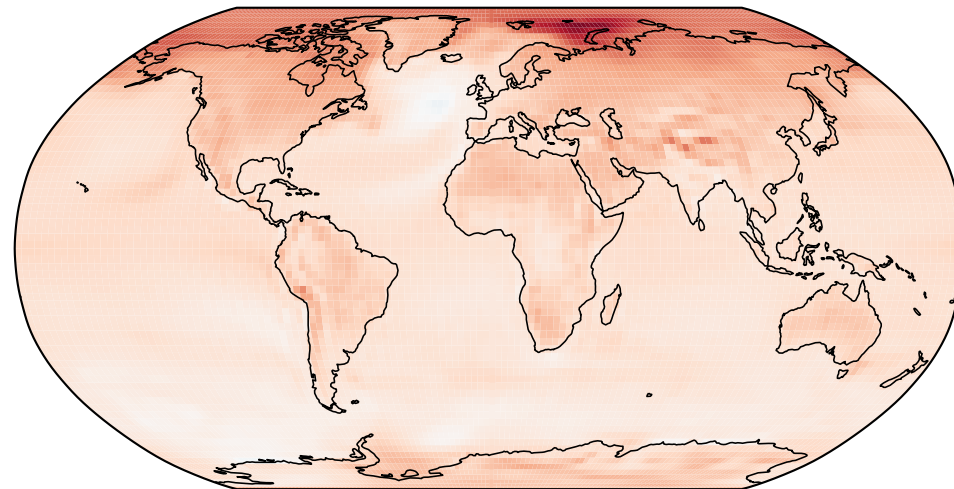
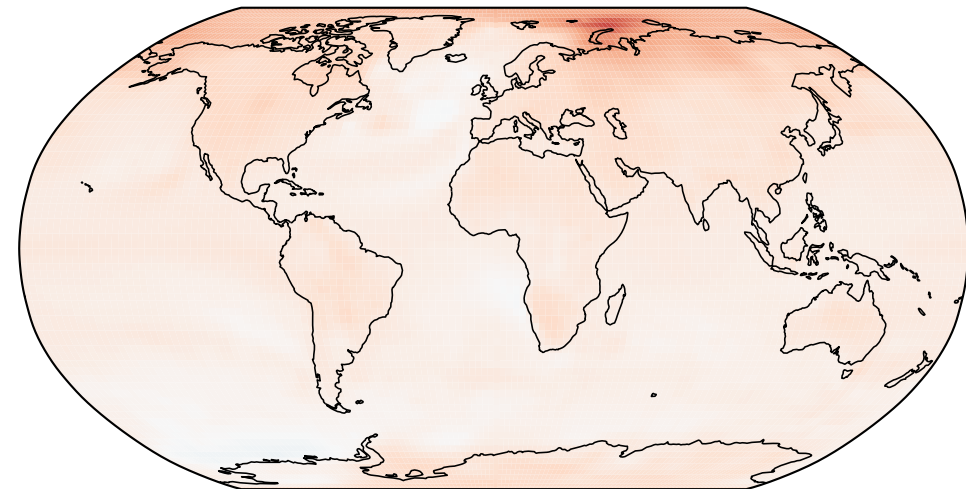
The Arctic time series highlights what is mildly surprising: even at 0.46 K piControl interannual standard deviation, the forced trajectories of r1 and r25 remain visually indistinguishable until roughly the mid-21st century, after which their year-to-year paths diverge by several Kelvin while sharing the same trend - a clean illustration of the signal-to-noise problem at regional scales.

MPI-ESM1-2-LR | near-surface air temperature anomaly (K)

future = mean(2091-2100); recent baseline = mean(2016-2025); preindustrial baseline = full piControl mean

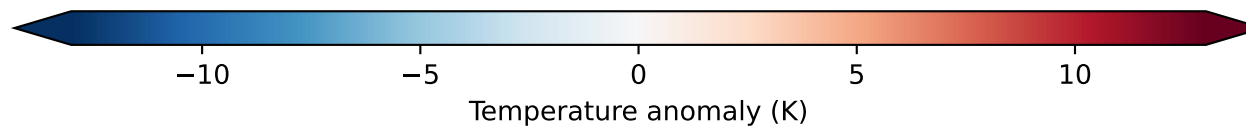
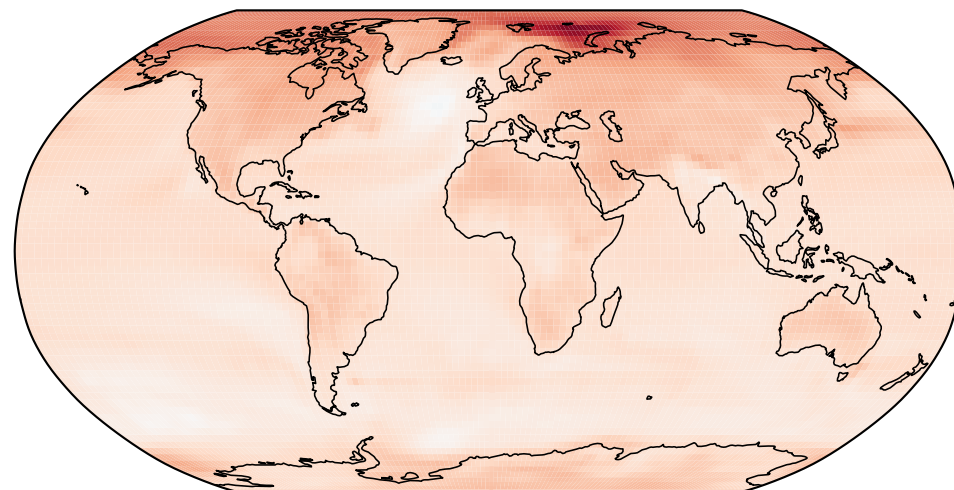
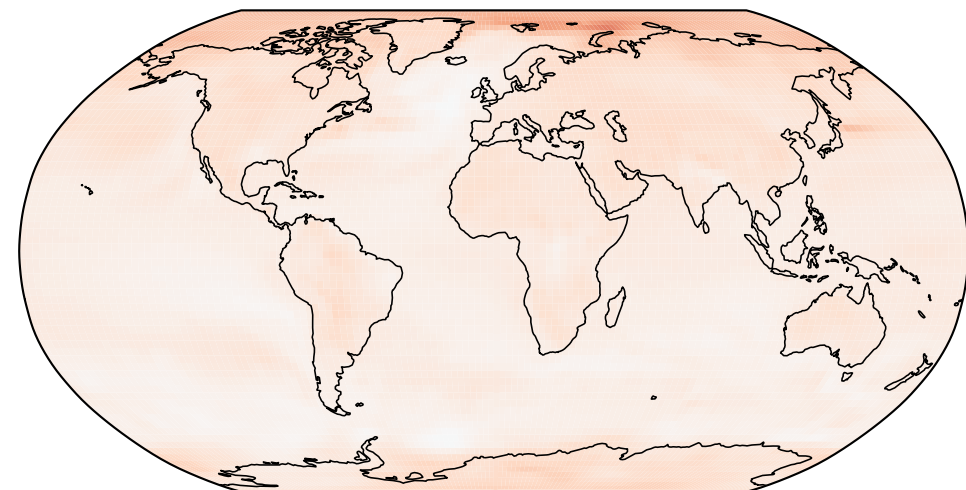
A: 2091-2100 minus 2016-2025

A: 2091-2100 minus piControl

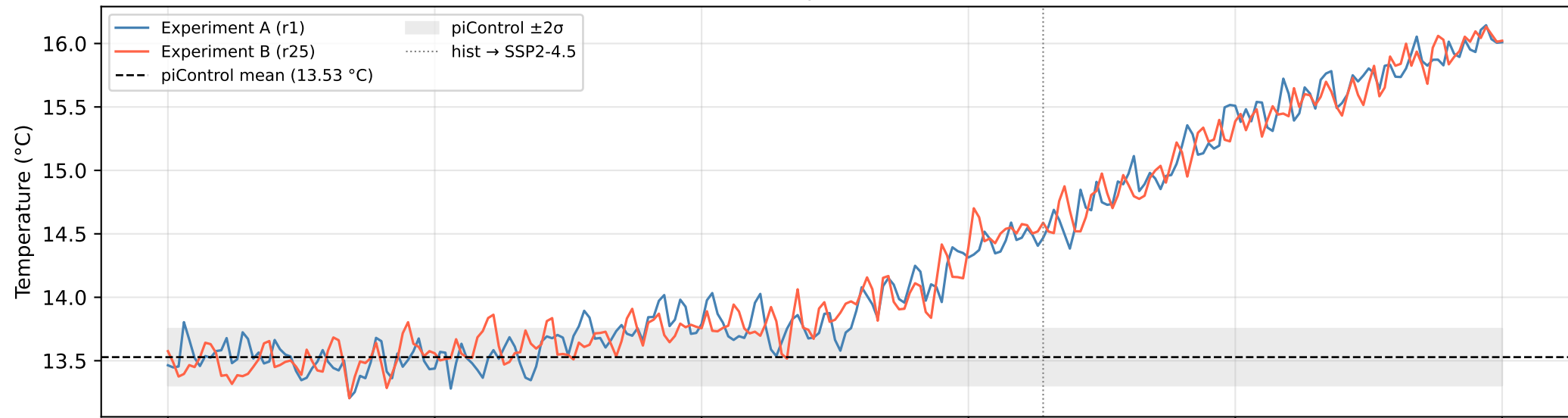


B: 2091-2100 minus 2016-2025

B: 2091-2100 minus piControl



Global mean | annual-mean tas



Arctic (60-90N) | annual-mean tas

